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University of Applied Sciences

KFT16480 Introduction to Autonomous Driving

Prof. Dr.-Ing. Rick Voßwinkel

Design and Implementation of a Path Following Control Algorithm

Presented by

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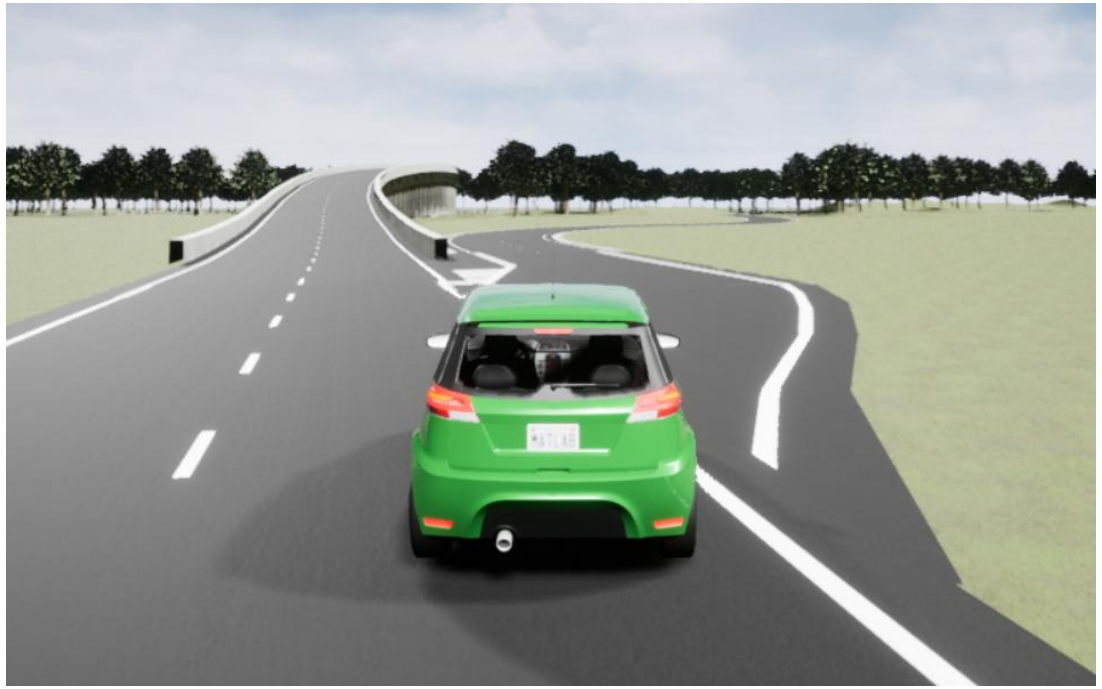
1. Vehicle Path Following

What is it?
The ability of a vehicle to **autonomously execute** a predefined trajectory with millimeter precision.
It bridges the gap between **Planning** and **Motion**.

Why it's Critical?
Autonomous Driving: Ensures the car stays centered in lane and navigates complex turns.
ADAS: Powers Emergency Lane Keeping and Adaptive Cruise Control to prevent collisions.

The Two Pillars of Control
Lateral (Steering): Controlling the front wheel angle to minimize path deviation.
Longitudinal (Speed): Managing throttle and braking to maintain the target velocity profile.



Source: MathWorks Student Competitions Team, *Vehicle Path Tracking Using Stanley Controller*, MathWorks File Exchange.

2. Path-Following Control Strategies

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Pure Pursuit

- Calculates the steering angle required to move the vehicle from its current position to a Look-ahead Point on the path.
- Like a person following a carrot on a stick.
- Highly robust and stable at low to medium speeds.

Stanley Controller

- The official path-following algorithm of the Stanford Racing Team. It uses nonlinear feedback based on:
- **Cross-track Error:** Distance from the front axle to the nearest path point.
- **Heading Error:** Angle difference between vehicle and path.

Model Predictive Control - MPC

- Treats path following as a constrained optimization problem over a future time horizon.
- Predicts future vehicle states and solves for the "best" steering/acceleration sequence while respecting physical limits (e.g., tire friction).

3. Why not Pure Pursuit or MPC?

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Pure Pursuit

- Pure Pursuit does not directly consider cross-track error or heading error
- Small look-ahead leads to oscillations. Large look-ahead results poor tracking in curves
- At higher speeds, Pure Pursuit can produce aggressive steering commands

MPC

- To work accurately, it requires a high-fidelity dynamic model of the vehicle, including mass, moment of inertia, and tire cornering stiffness
- MPC solves an online optimization problem (usually Quadratic Programming) at every single time step.
- This requires significant CPU power and can introduce **latency**.

While MPC offers predictive capabilities, we opted for the Stanley Controller because of its ***geometric robustness*** *and* ***computational efficiency****.*

4. Stanley Controller

The Stanley controller is a geometric **lateral control** algorithm used for vehicle path tracking.

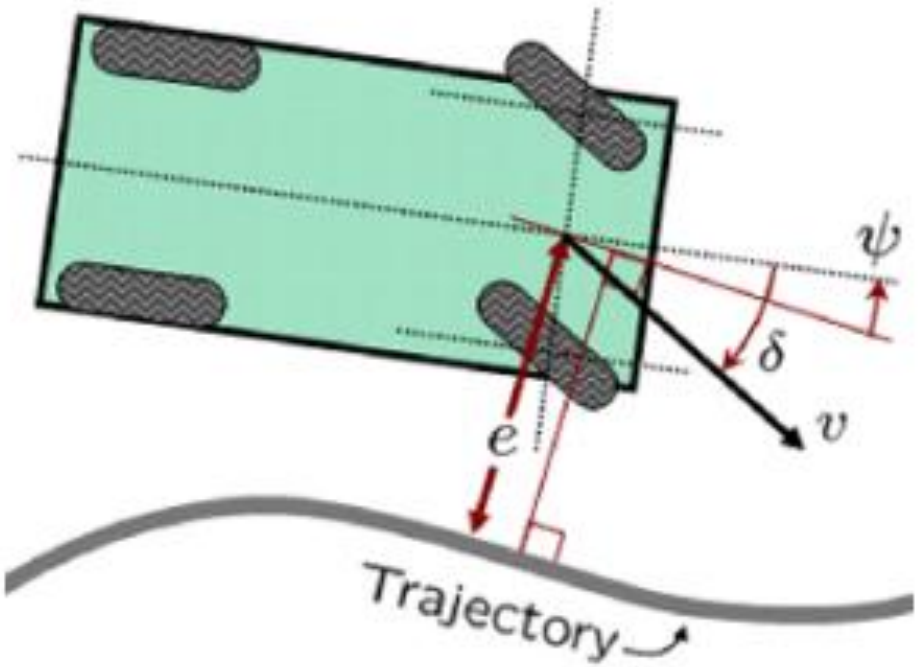
It computes the steering angle based on two key errors:

1. Heading Error (ψ)

- Difference between the vehicle's current yaw angle and the direction of the reference path.
- Ensures the vehicle is oriented along the path.

2. Cross-Track Error (*e*)

- Lateral distance between the vehicle position and the reference path.
- Ensures the vehicle converges back to the path

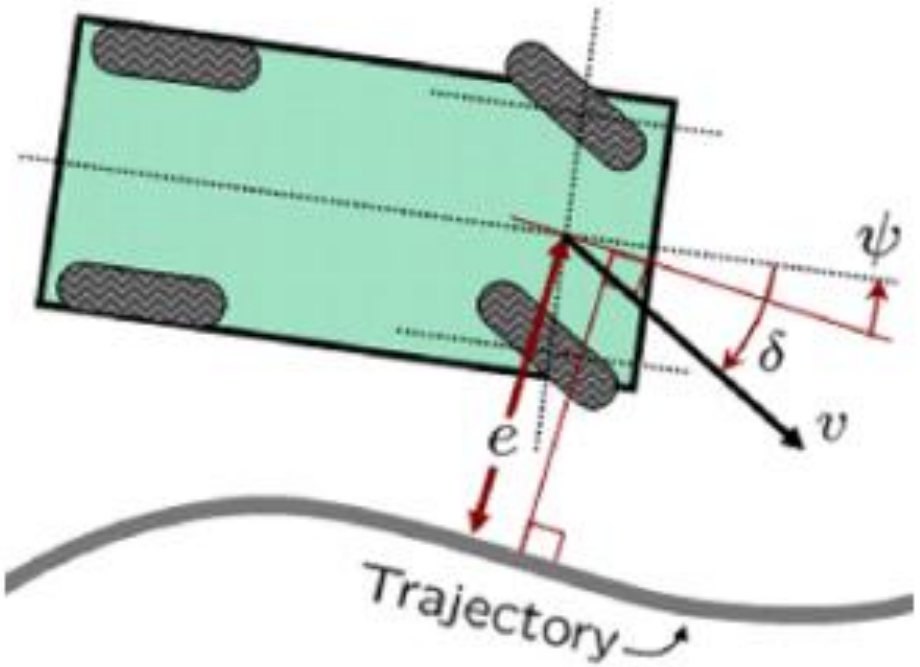


Source: G. M. Hoffmann et al., 'Autonomous Automobile Trajectory Tracking for Off-Road Driving,' 2007.

4. Stanley Controller

$$\delta_v = \psi_e + \arctan\left(\frac{k.e}{v + v_{comp}}\right)$$

- $\delta_v = Steering\ Angle$
- $\psi_e = Heading\ Error$
- $e = Cross\ Track\ Error$
- $k = Gain\ Parameter$
- $v = Velocity$
- $v_{comp} = velocity\ compensation$



Source: G. M. Hoffmann et al., 'Autonomous Automobile Trajectory Tracking for Off-Road Driving,' 2007.

Tuning of the Controller

Lateral Dynamic:
K Gain
V Comp

$$\delta_v = \psi_e + \arctan\left(\frac{k.e}{\boxed{v + v_{comp}}}\right)$$

Increase in denominator tends to lower the Steering Angle

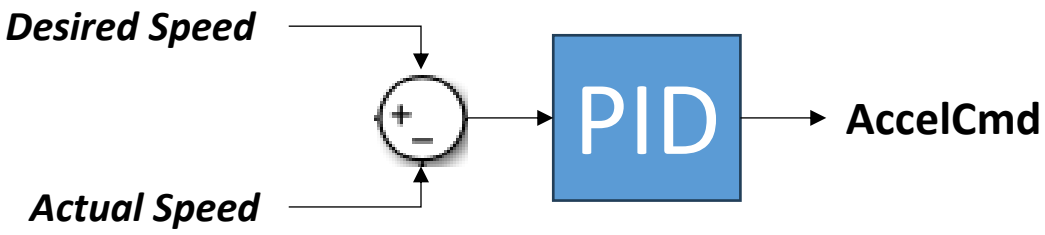
5. PID for Longitudinal Control

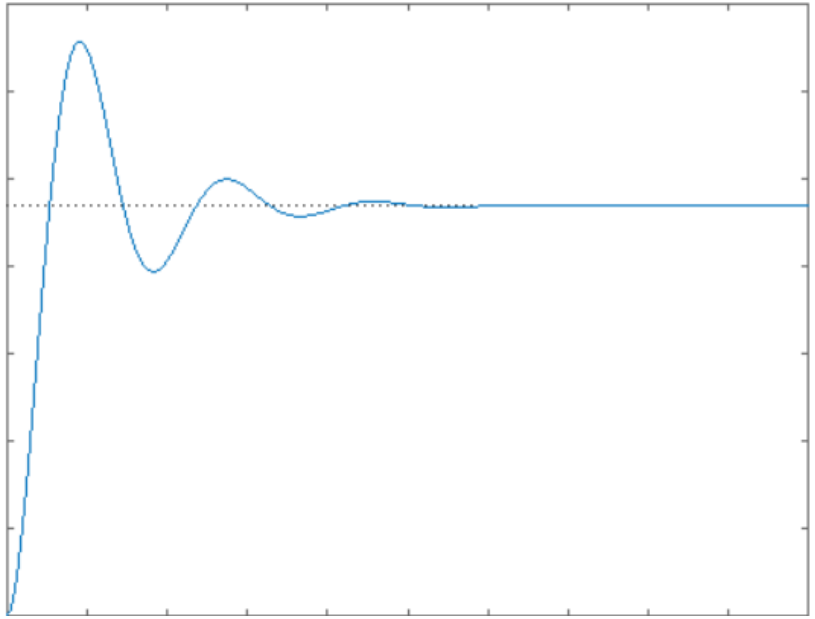
- A PID controller is used to control vehicle acceleration, due to easiness and widely applied in automotive longitudinal control.
- **Input**: Speed error (difference between desired and actual speed)
- **Output**: Longitudinal acceleration command
- Longitudinal Dynamics

  K_P - Reacts to current speed error

  K_I - Eliminates steady-state speed error

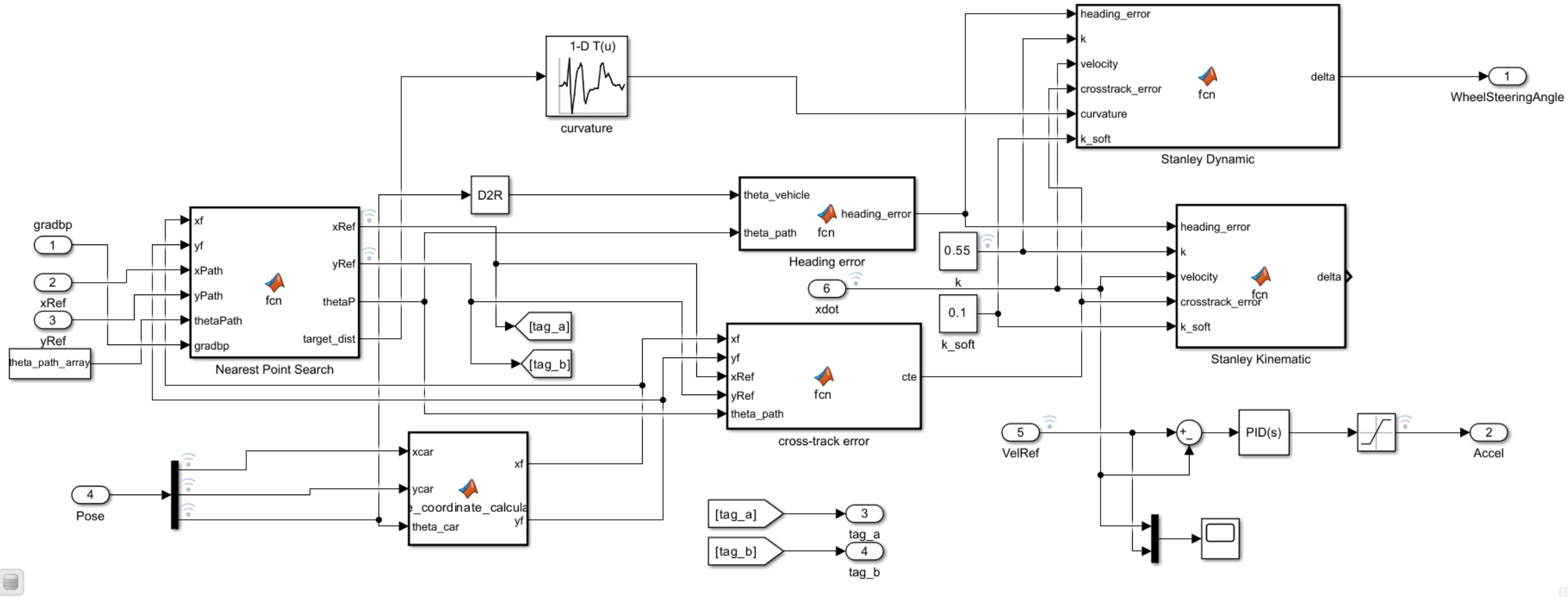
  K_D - Improves smoothness and stability
- Optimizing involved finding a good balance between these tunable variables and desired output.





Messner, et. al (2023). *Control Tutorials for MATLAB and Simulink - PID Control. University of Michigan & Carnegie Mellon University.*

6. Controller Overview in MATLAB



Snapshot of the Controller from the MATLAB model

7. Optimization

Initially we used the built-in optimization algorithm (SQP) to explore the minimum error parameters

- Scale = Step towards minima
- Minimum and Maximum Boundary requirements

Create Design Variables set: DesignVars

	Continuous Variable	Value	Minimum	Maximum	Scale
☑	K_Stanley	0.8	0.1	10	1
☑	Ki_long	0.5	0	5	0.5
☑	Kp_long	3	0.1	10	4
☑	K_soft	0.8	0.1	3	1

Drawback

Every step of the optimizer may have violated boundary conditions and hence the change of parameters didn't make much sense

✓ Due to constraint violations and inconsistent convergence, manual tuning was preferred for final controller parameters.

7. Optimization

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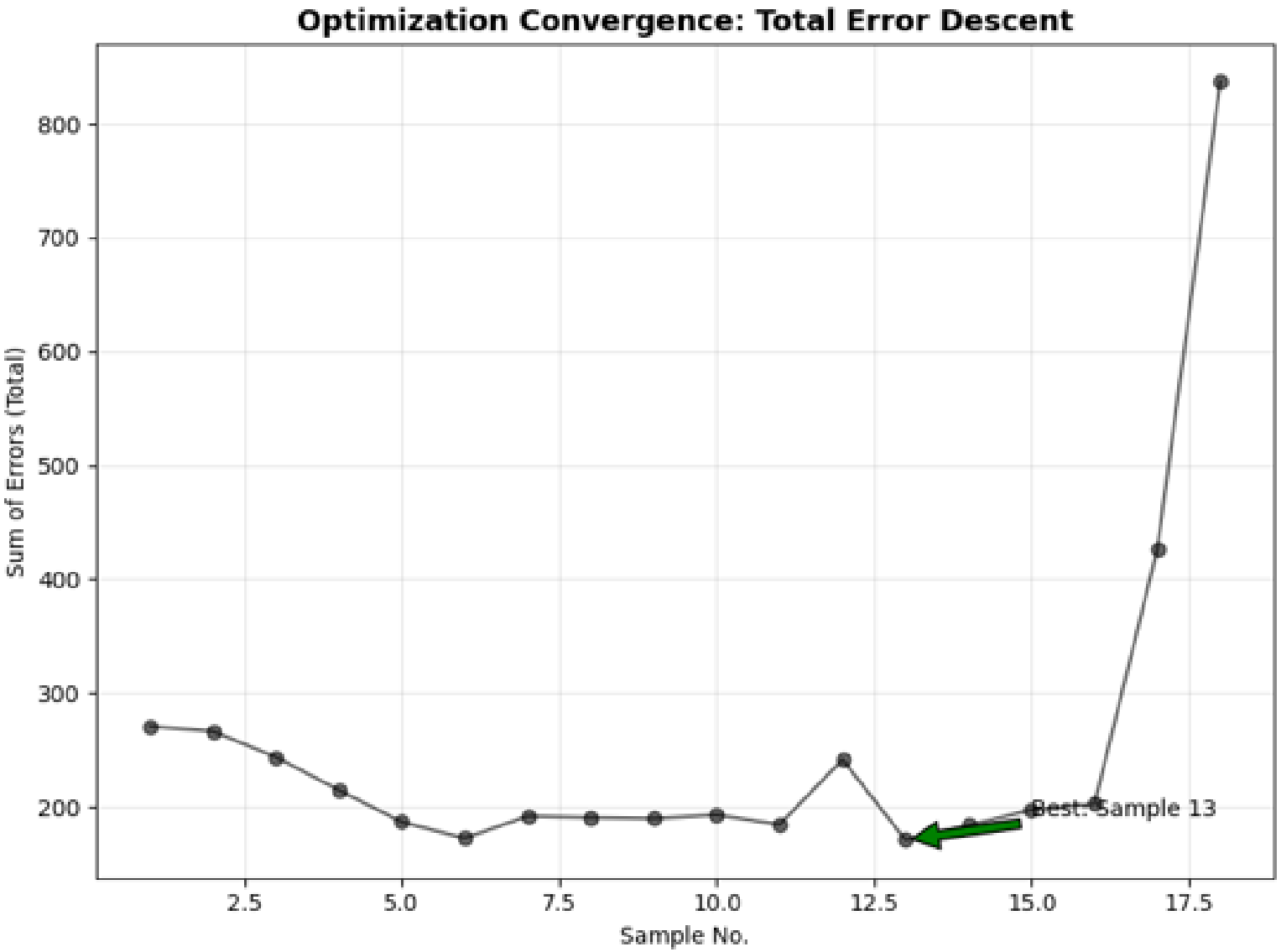
Controller Parameter Tuning (Manual)

Sample	Stanley Gain (K)	Softening Gain (K_s)	Wheelbase (L)	P	I	D	N	Int. Sat Low	Int. Sat High	Ext. Sat Low	Ext. Sat High	**Velocity Error**	**Path Error**	Total Error
1	0.4	0.1	3	3	0.5	0.1	100	-3	3	-5	3	83.6	187.0	270.6
2	0.4	0.1	3	3	0.5	0.1	100	-3	3	-5	3	85.47	180.8	266.27
3	0.5	0.5	3	3	0.5	0.1	100	-3	3	-5	3	83.37	160.2	243.57
4	0.6	0.6	3	3	0.5	0.1	100	-3	3	-5	3	81.52	133.0	214.52
5	0.7	0.7	3	3	0.5	0.1	100	-3	3	-5	3	90.98	95.49	186.47
6	0.8	0.8	3	3	0.5	0.1	100	-3	3	-5	3	**85.32**	**86.96**	**172.28**
7	0.9	0.9	3	3	0.5	0.1	100	-3	3	-5	3	111.7	80.03	191.73
8	1.0	1.0	3	3	0.5	0.1	100	-3	3	-5	3	123.6	66.85	190.45
9	1.1	1.1	3	3	0.5	0.1	100	-3	3	-5	3	134.9	54.73	189.63
10	1.2	1.2	3	3	0.5	0.1	100	-3	3	-5	3	140.2	52.63	192.83
11	1.3	1.3	3	3	0.5	0.1	100	-3	3	-5	3	142.9	41.42	184.32
12	1.4	1.4	3	3	0.5	0.1	100	-3	3	-5	3	191.5	49.87	241.37
13	1.5	1.5	3	3	0.5	0.1	100	-3	3	-5	3	136.3	34.46	170.76
14	1.6	1.6	3	3	0.5	0.1	100	-3	3	-5	3	137.9	45.72	183.62
15	1.7	1.7	3	3	0.5	0.1	100	-3	3	-5	3	160.6	36.57	197.17
16	1.1781	0.1	3	2.5	1.0	0.1	100	-3	3	-5	3	161.4	40.92	202.32
17	0.45	0.1	3	0.4	0.02	0.3	100	-3	3	-5	3	291.2	134.9	426.10
18	0.55	0.1	3	3	0.5	0.1	100	-5	2	-5	1	**781.2**	**56.04**	**837.24**

7. Optimization



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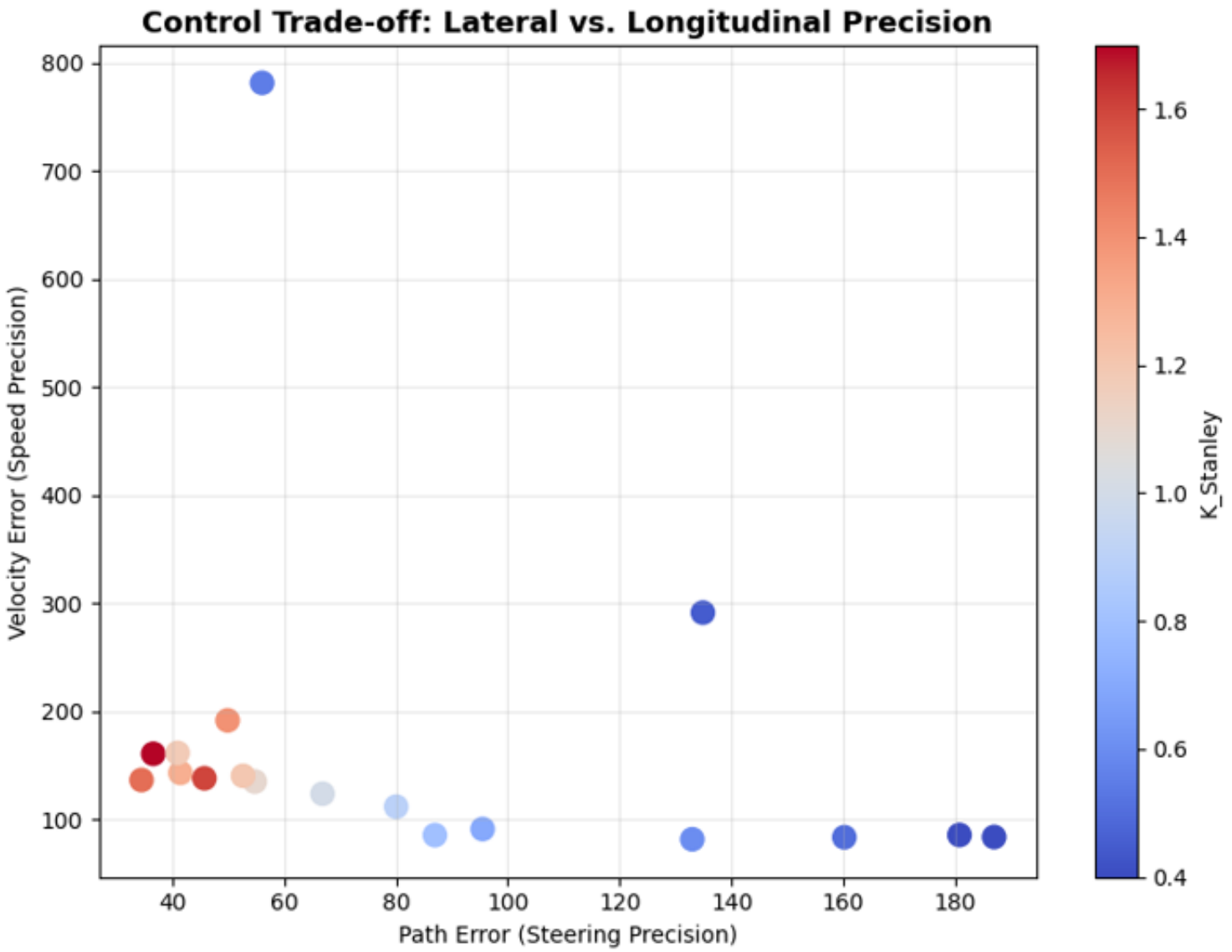

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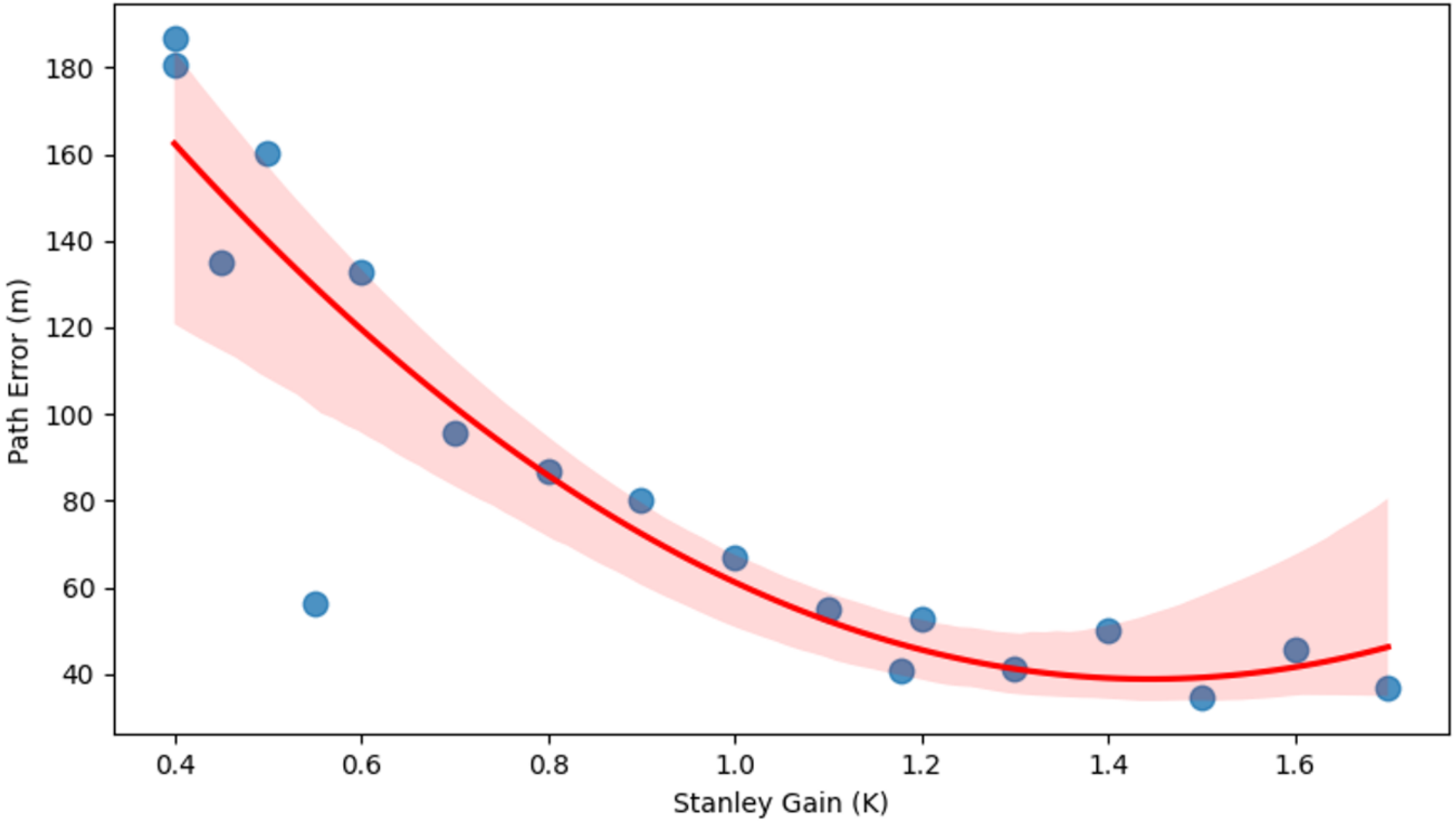
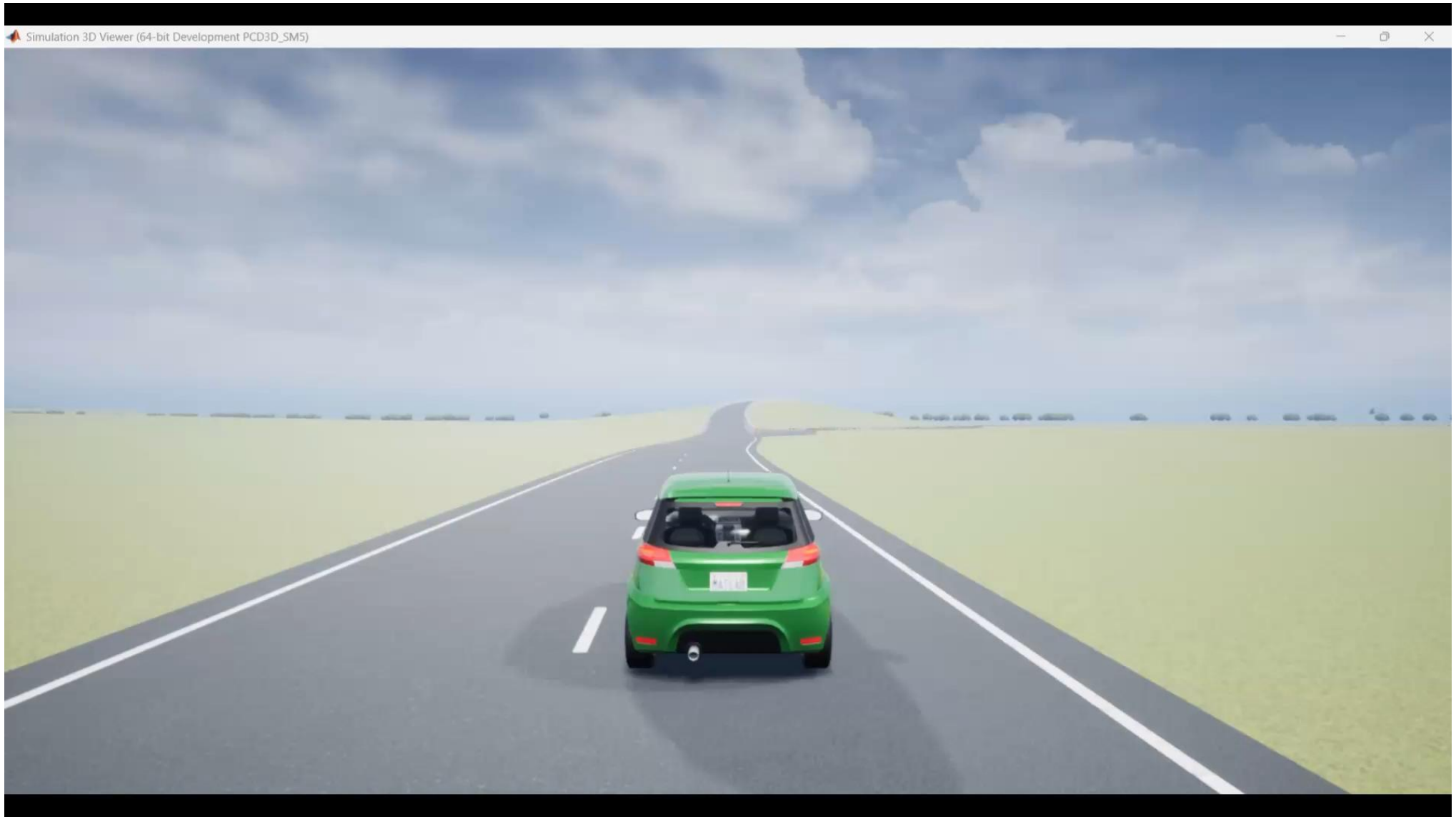
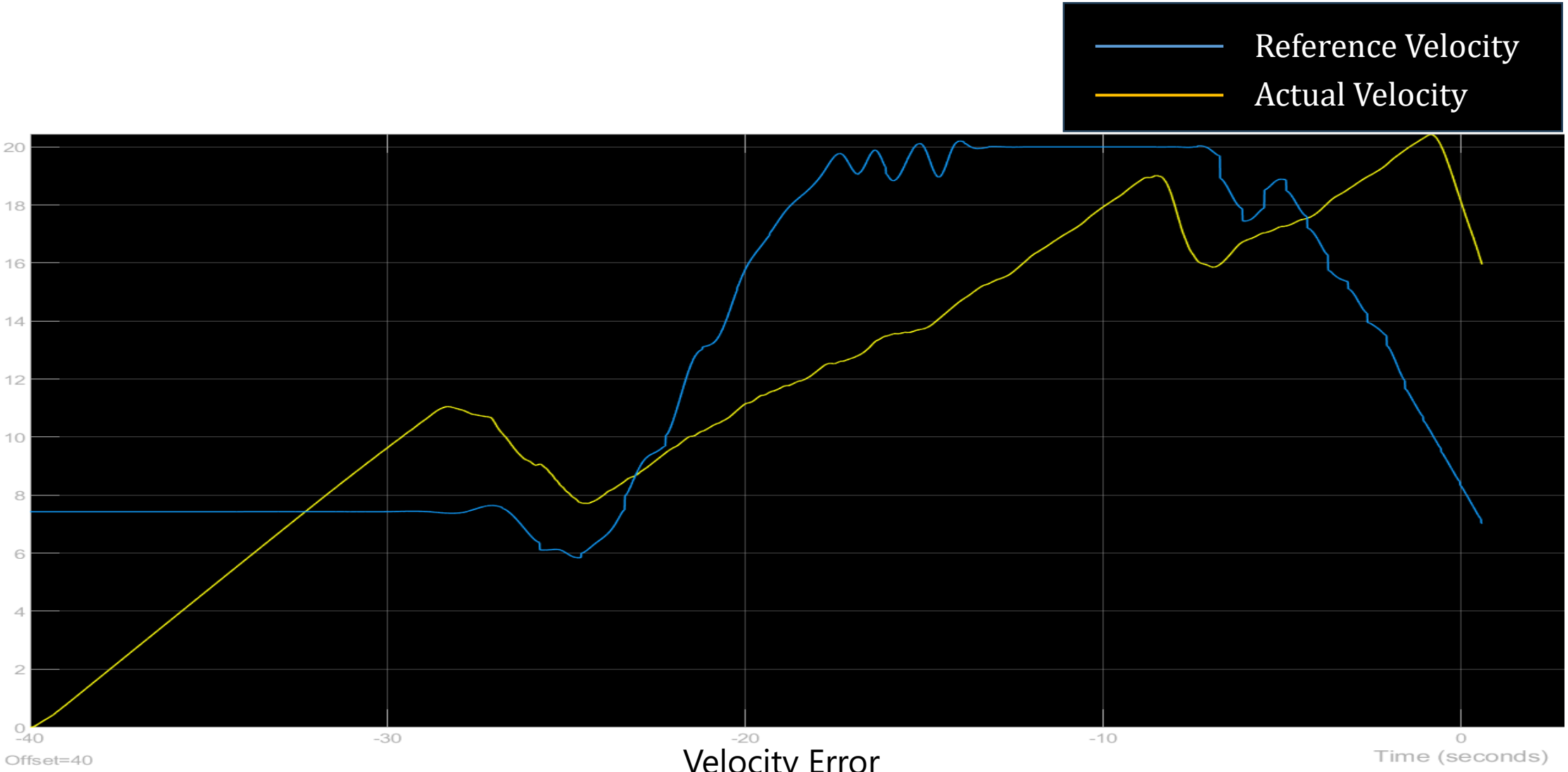


8. Simulation

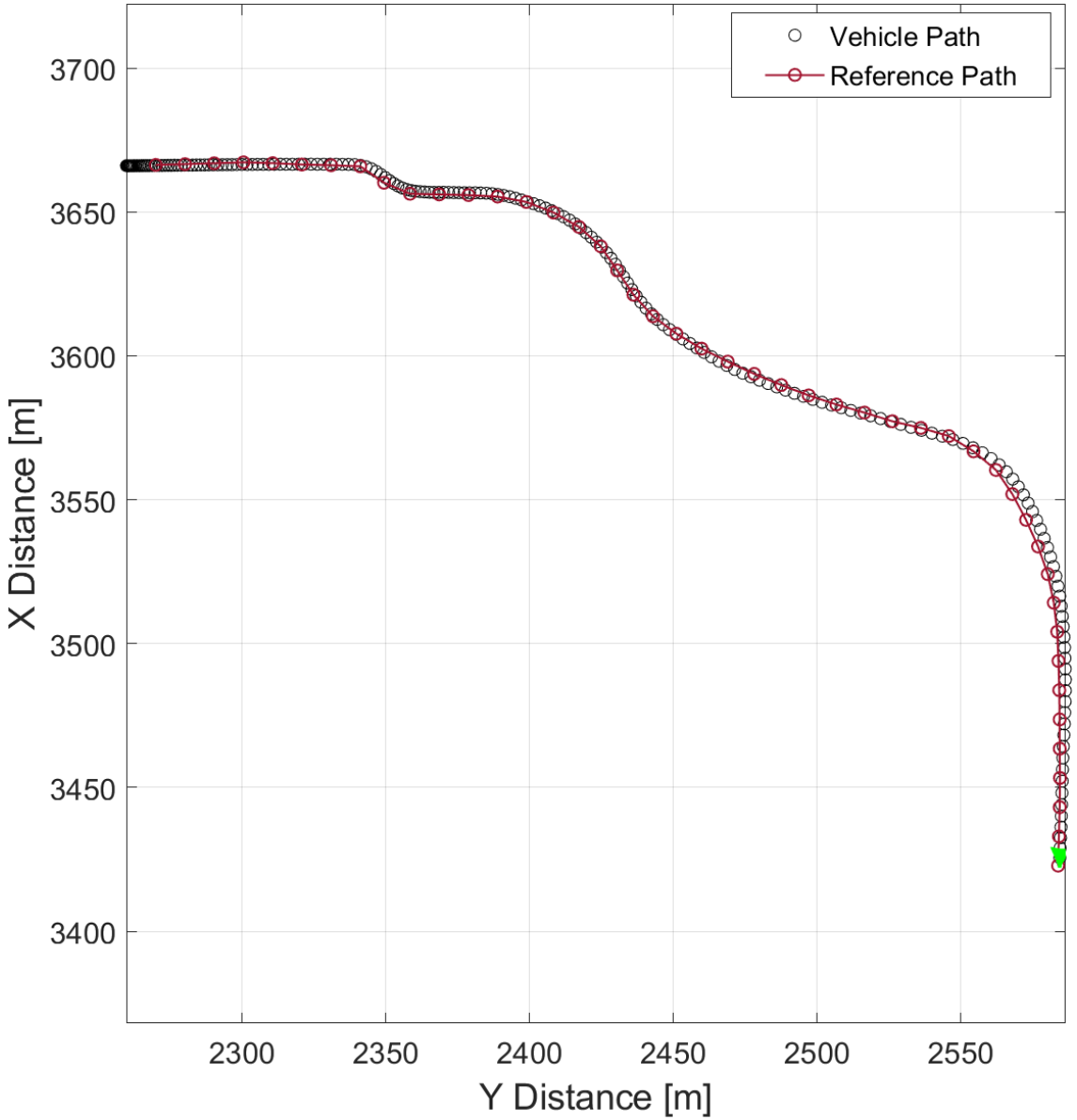


9. Results



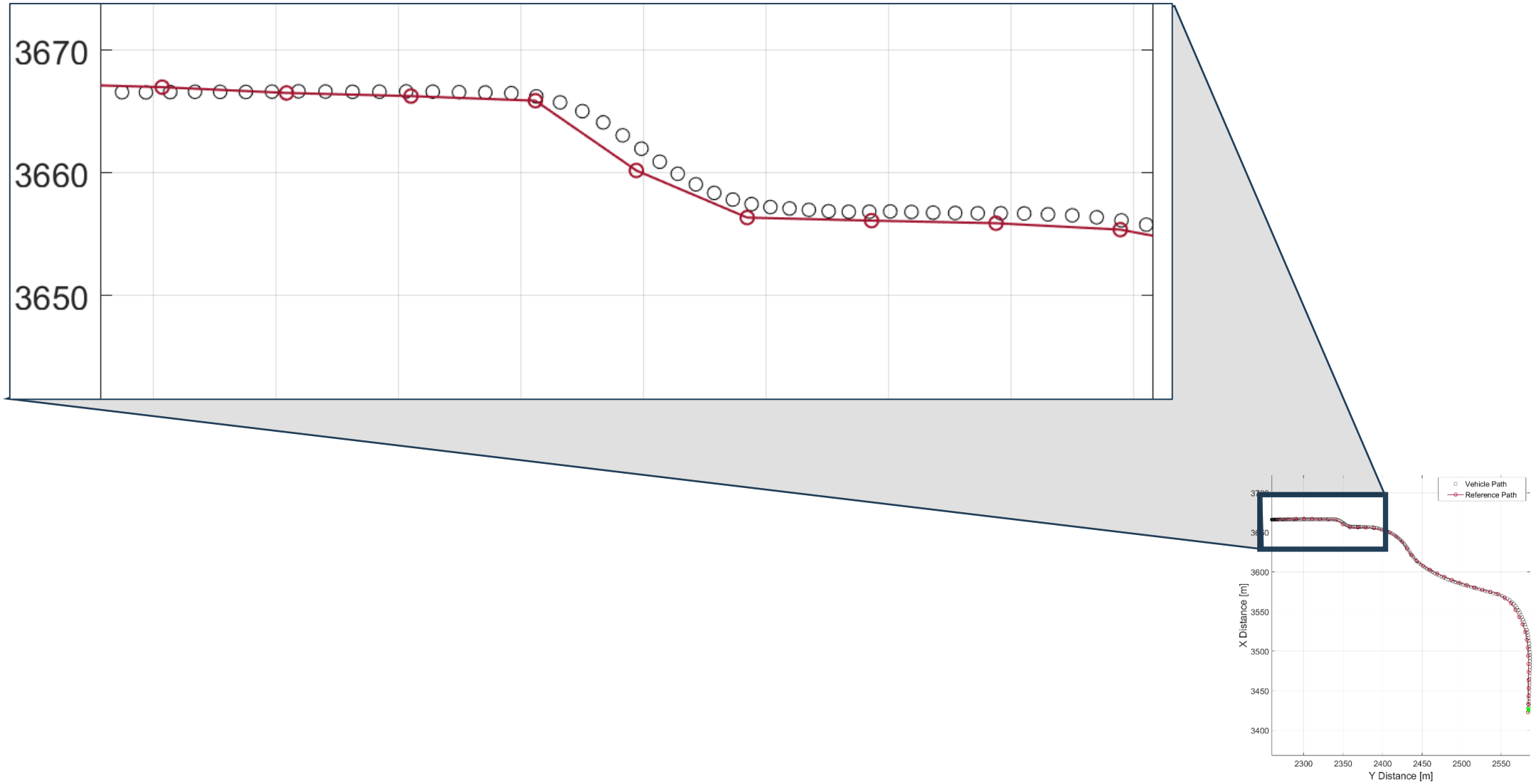
Velocity Error

9. Results


Vehicle Path
Reference Path
3700
3650
3600
3550
3500
3450
3400
X Distance [m]
2300
2350
2400
2450
2500
2550
Y Distance [m]

9. Results

Close up view of First Curve



10. Conclusion

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- A complete vehicle path-following control system was successfully designed and implemented in MATLAB/Simulink.
- Different path-following strategies were studied and compared.
- **Pure Pursuit** was found to be simple but sensitive to tuning and less stable at higher speeds.
- **Model Predictive Control (MPC)** offers strong predictive performance but was not selected due to high computational complexity.
- The **Stanley Controller** was chosen for lateral control due to:
  - ✓ Explicit handling of cross-track and heading errors
  - ✓ Improved stability and robustness
  - ✓ Low computational cost
- A **PID controller** was selected for longitudinal control due to its simplicity, reliability, and wide industrial adoption.
- Simulation results demonstrate accurate path tracking and stable velocity control.

References

- **Hoffmann, G. M., Tomlin, C. J., Thrun, S., & Levinson, J. B. (2007).** *Autonomous Automobile Trajectory Tracking for Off-Road Driving: Controller Design, Experimental Validation and Racing*. Stanford University, Stanford, CA. (The original source for the Stanley Control Law).

- **Messner, W., Tilbury, D., Shahian, B., & Rohani, K. (2023).** *Control Tutorials for MATLAB and Simulink - PID Control*. University of Michigan & Carnegie Mellon University. Available at: https://ctms.engin.umich.edu/.

- **MathWorks Student Competitions Team (2021)**. *Vehicle Path Tracking Using Stanley Controller*. https://www.mathworks.com/matlabcentral/fileexchange/88977-vehicle-path-tracking-using-stanley-controller